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Graphing Inequalities

Each inequality is a line and a side. Draw the lines (solid or dashed), test a point, and the region is where every condition is true.

THE TOPIC ON ONE MAP

Graphing Inequalities

01 — Shade the region from the inequalities

draw $x = 1.5$, $y = x$, $x + y = 6$ solid or dashed

shade the unwanted parts

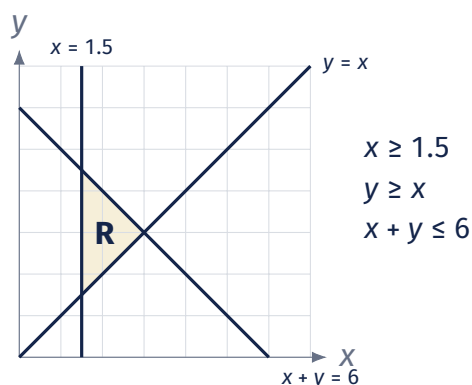
02 — Write the inequalities for a shaded region

three lines, three inequalities find each line's equation

03 — Integer points in a region

mark every point with a cross list the possible values

PICTURE IT Three lines, one region:



01 Shade the Region from the Inequalities

Draw every boundary line first. Then test a point for each inequality to find its side. R is the part where all of them are true — label it.

WHEN YOU SEE → YOU DO

“draw and label the line $x = 1.5$ ”

→ vertical line; $y = 2$ is horizontal

$y \leq x + 1$, “ $x + y \leq 7$ ”

→ draw the line from two points

< or >

→ dashed line

“by shading the unwanted regions”

→ shade the parts that are **not** wanted; R is left clear

“label the region R ”

→ write R inside it

LINE AND SIDE The sign decides two things:

ZONE THE LINE

$\leq$ or $\geq$: **solid** < or >: **dashed**

a dashed line is not part of the region

ZONE THE SIDE

test a point, e.g. (0, 0)

true → that side is in; false → the other side

Draw each line

Solid or dashed

Test a point

Keep the overlap

Label R

WORKED MODEL • JUNE 2022 P2H

Shade the region $y \geq 1$, $x \geq 2$, $x + y \leq 7$.

1 draw $y = 1$, $x = 2$, and $x + y = 7$ through (0, 7) and (7, 0)

lines

2 (3, 2): $2 \geq 1$ yes, $3 \geq 2$ yes, $5 \leq 7$ yes

test a point inside

ANSWER triangle with corners (2, 1), (6, 1), (2, 5)

× **TRAP** Testing (0, 0) when it lies on a line ($y = x$). Pick another point, like (1, 0).

IDEA 01

Practise: Shade the Region from the Inequalities

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Shade

CH 3 · TOPIC 8 3 marks

Shade the region $x \geq 1$, $y \geq 2$, $x + y \leq 6$. Give the coordinates of its corners.

SELECTED PRACTICE

Q2**PRACTICE**

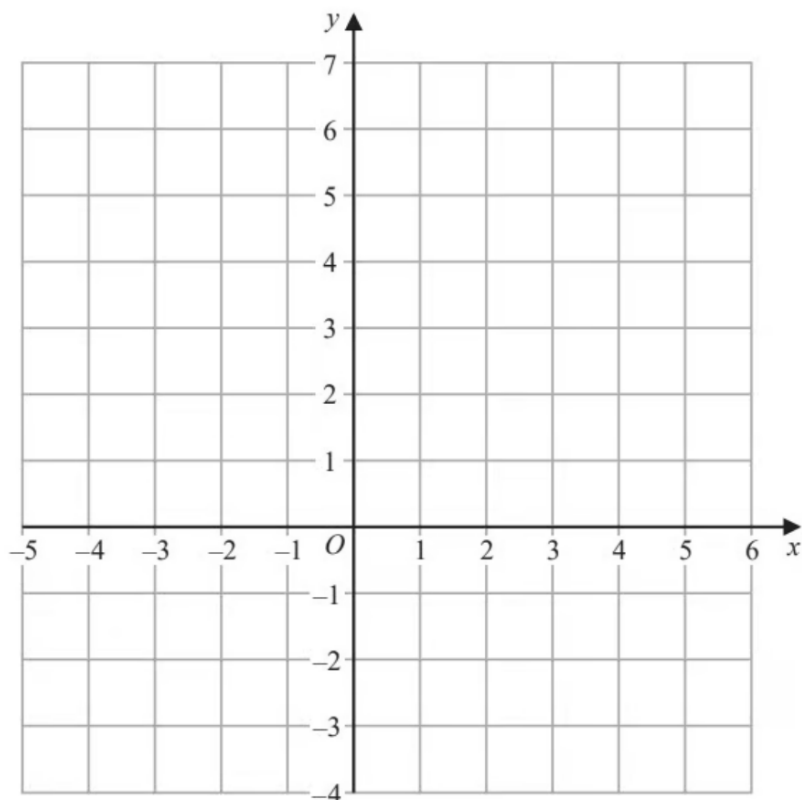
Practice · Easy

CH 3 · TOPIC 8 3 marks

- 2 Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \leq 4 \quad \text{and} \quad y \geq -2 \quad \text{and} \quad y \leq x$$

Label the region **R**.

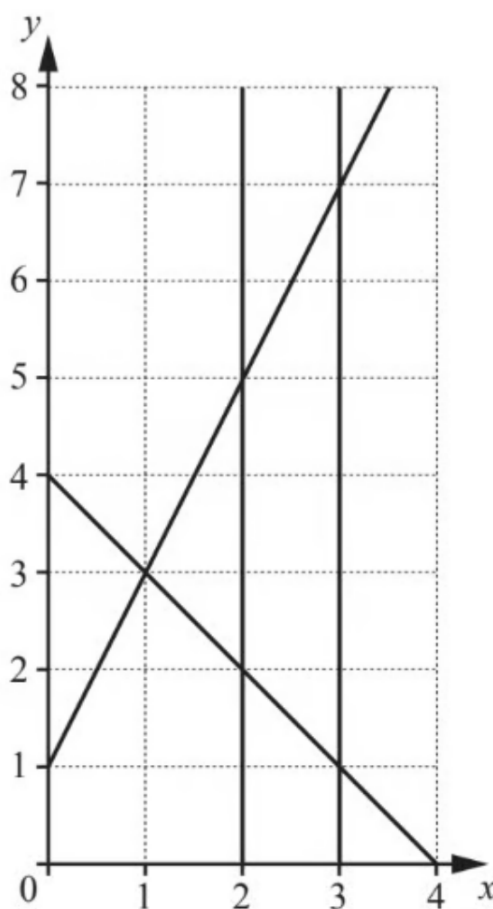


(3 marks)

Q3**PRACTICE**

Practice • Easy

CH 3 • TOPIC 8

3 marks**10**

By shading the **unwanted** regions of the grid, find and label the region R that satisfies the following four inequalities.

$x \leq 3$

$x \geq 2$

$y \leq 2x + 1$

$y \geq 4 - x$

(3 marks)

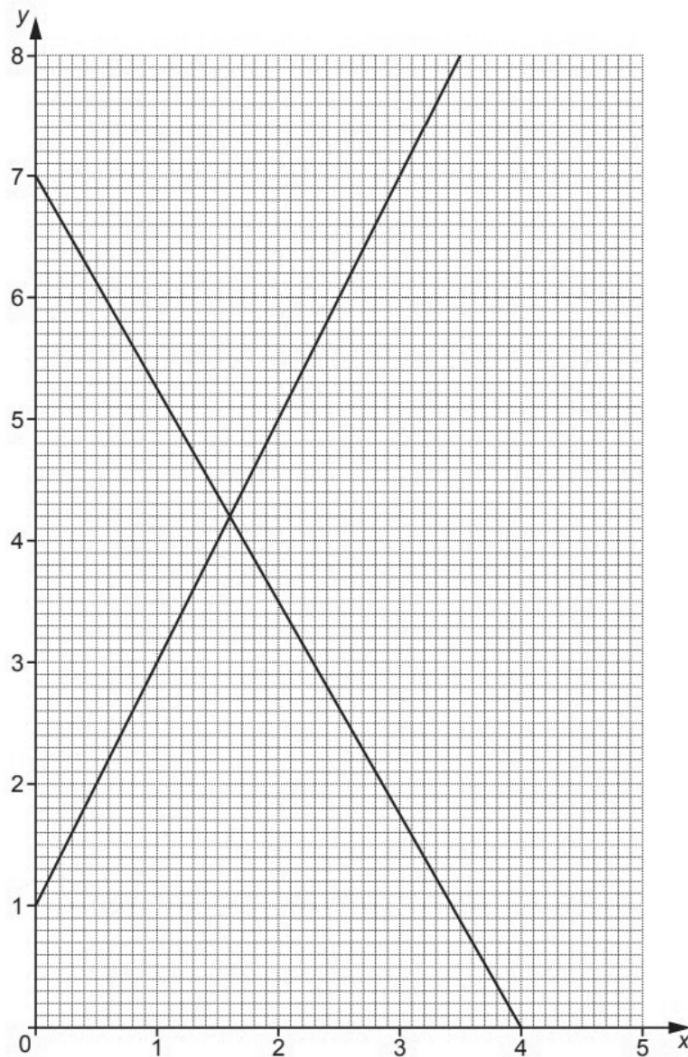
Q4**PRACTICE**

Practice · Medium

CH 3 · TOPIC 8

5 marks

10 The diagram shows the lines $y = 2x + 1$ and $7x + 4y = 28$



The region R satisfies these inequalities.

$$y \leq 2x + 1 \quad 7x + 4y \geq 28 \quad y > 1$$

By drawing a third straight line, find and label the region R that satisfies these inequalities.

(5 marks)

Q5**PRACTICE**

Practice • Hard

CH 3 • TOPIC 8 6 marks

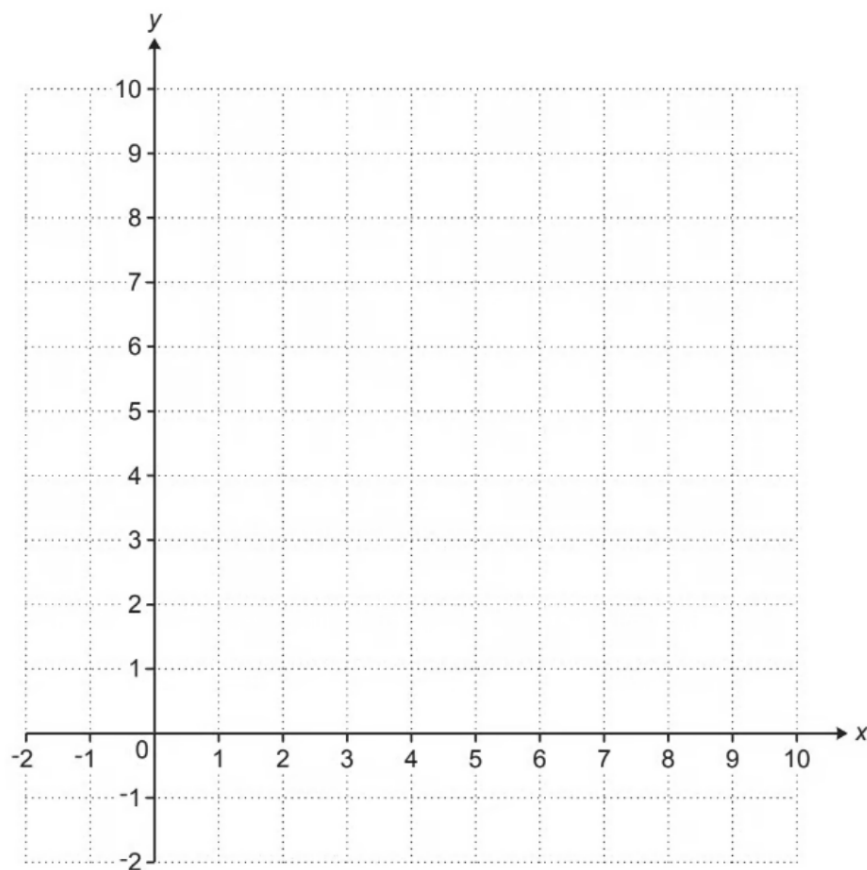
8 Region **R** satisfies these inequalities.

$$y > 3$$

$$y \geq x$$

$$x + y \leq 9$$

By drawing three straight lines on the grid, find and label the region **R**.



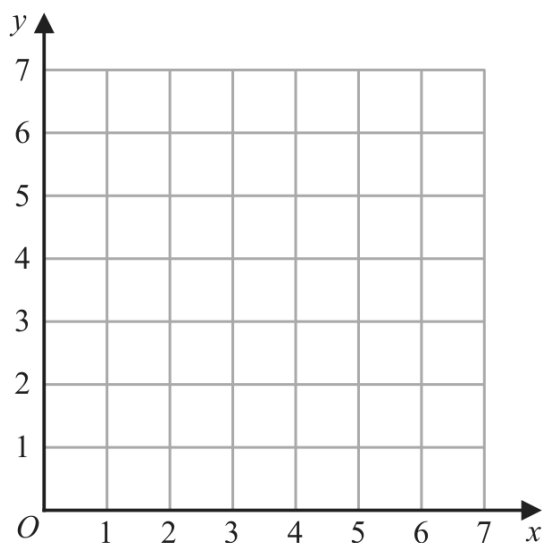
(6 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 2H · Q4

CH 3 · TOPIC 8 4 marks

4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$$x \geq 1.5$$

$$y \geq x$$

$$x + y \leq 6$$

Label the region **R**.

(Total for Question 4 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q7

CH 3 · TOPIC 8 5 marks

- 7 (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates $(0, 5)$

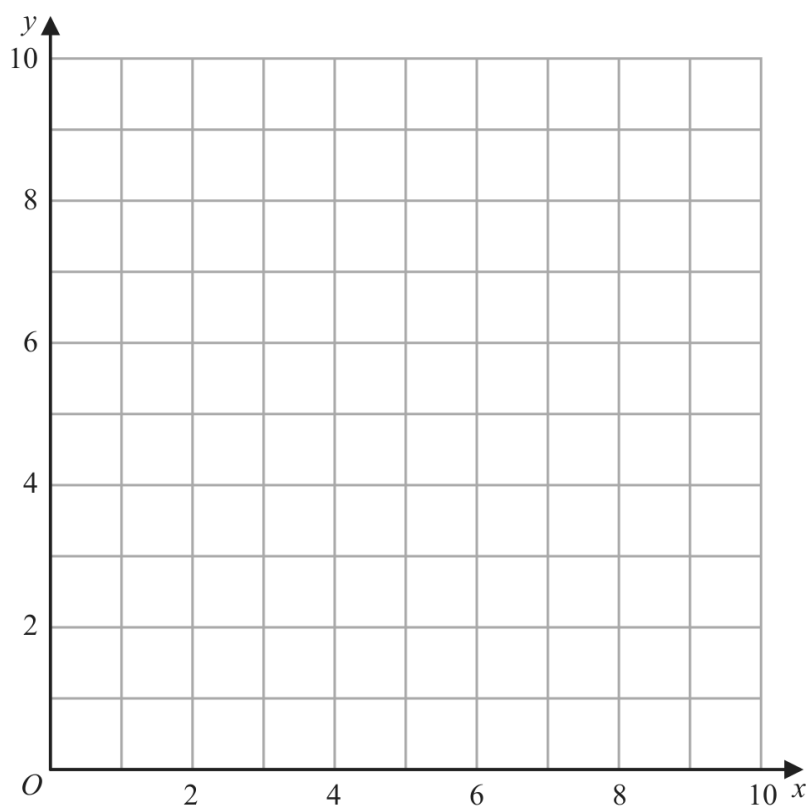
- (b) Show, by shading on the grid, the region defined by **all three** of the inequalities

$$x \leq 6$$

$$y \geq 2$$

$$y \leq x + 1$$

Label the region **R**



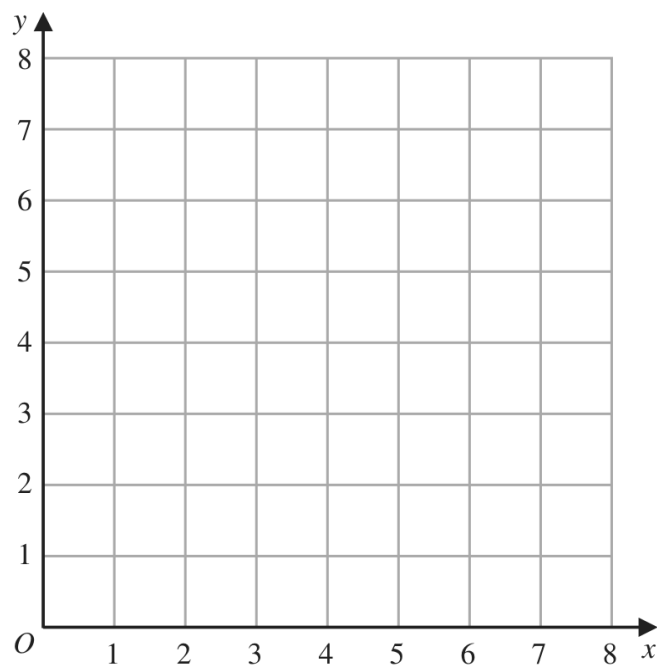
(Total for Question 7 is 5 marks)

Q8 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 2H · Q2

CH 3 · TOPIC 8 4 marks

2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$ $x \geq 2$ $x + y \leq 7$

Label the region **R**.

(Total for Question 2 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2H · Q2

CH 3 · TOPIC 8 4 marks

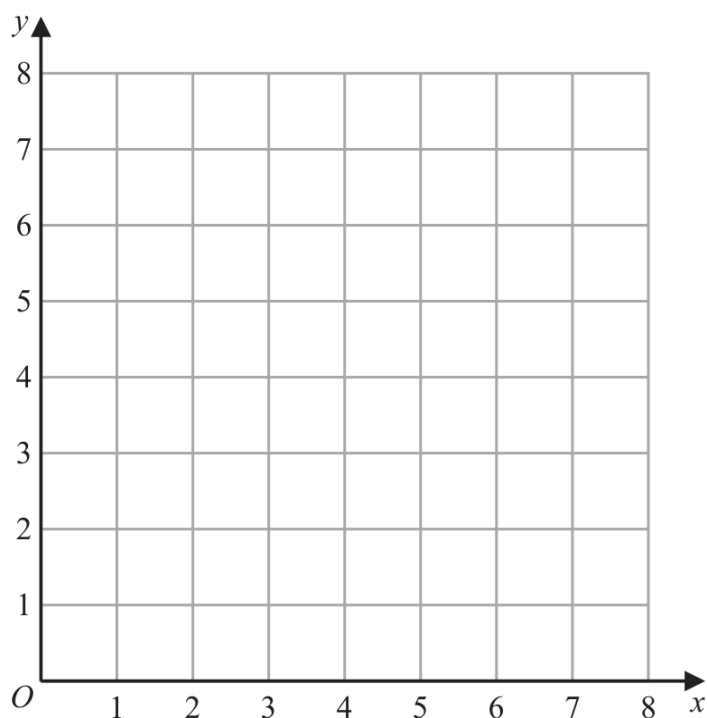
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x + 1$$

Label the region **R**

(Total for Question 2 is 4 marks)

Q10**PAST PAPER**

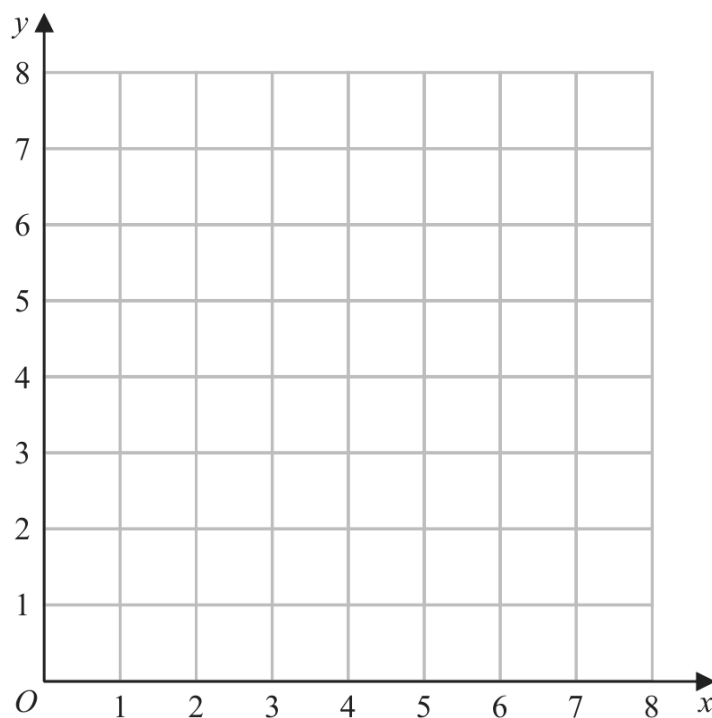
Edexcel 4MA1 · November 2024 · Paper 2H · Q7

CH 3 · TOPIC 8 4 marks

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geq 3 \quad y \geq 1 \quad x + y \leq 7$$

Label the region **R**

(Total for Question 7 is 4 marks)

02 Write the Inequalities for a Shaded Region

One inequality for each boundary line. Find the equation of each line, then decide the sign by testing a point inside the region.

WHEN YOU SEE → YOU DO

a vertical / horizontal boundary

→ $x \geq a$ / $y \leq b$

a sloping line

→ find $y = mx + c$ from the grid

“write down three inequalities”

→ three lines → three answers

a solid line

→ use $\leq$ or $\geq$ (unless told otherwise)

Name each line

Pick a point inside

Choose $\leq$ or $\geq$

WORKED MODEL • JANUARY 2022 P2H

The region is bounded by $x = -1$, $y = x$ and $x + 2y = 8$ (it lies left of $y = x$ and below $x + 2y = 8$).

1 test $(0, 2)$: $0 \geq -1$, $2 \geq 0$, $4 \leq 8$

a point inside

ANSWER $x \geq -1$, $y \geq x$, $x + 2y \leq 8$

IDEA 02

Practise: Write the Inequalities for a Shaded Region

My questions first, then selected practice, then real past papers.

Q11 DR ESLAM Write

CH 3 · TOPIC 8 3 marks

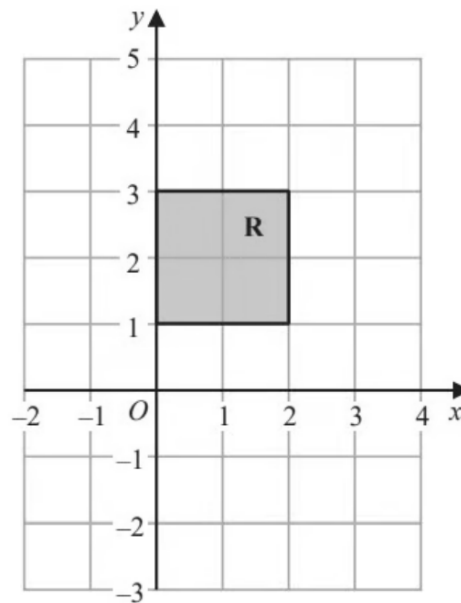
A triangle is bounded by $y = 0$, $x = 4$ and $y = x$, and lies below $y = x$. Write its three inequalities.

SELECTED PRACTICE

Q12 **PRACTICE** Practice · Easy

CH 3 · TOPIC 8 2 marks

3



The region **R**, shown shaded in the diagram, is bounded by four straight lines.

Write down the inequalities that define **R**.

(2 marks)

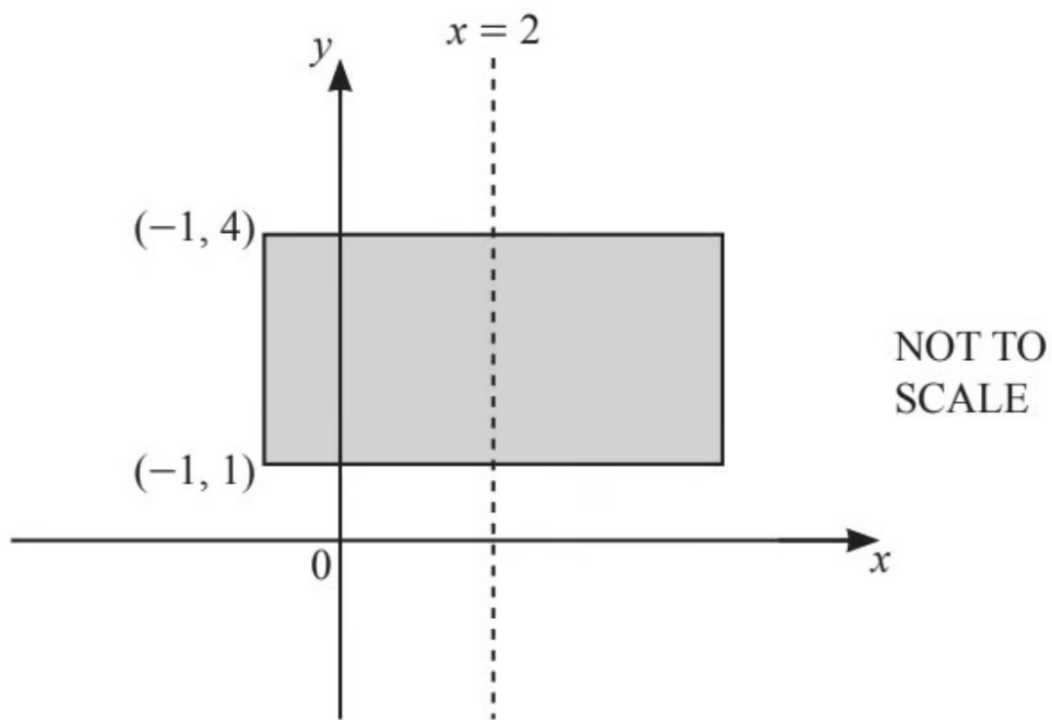
Q13**PRACTICE**

Practice · Easy

CH 3 · TOPIC 8

2 marks

8



The diagram shows a rectangle with a line of symmetry at $x = 2$. Two vertices of the rectangle are at $(-1, 1)$ and $(-1, 4)$.

The shaded region is defined by the inequalities $a \leq x \leq b$ and $c \leq y \leq d$.

Find the values of a , b , c and d .

$a =$

$b =$

$c =$

$d =$

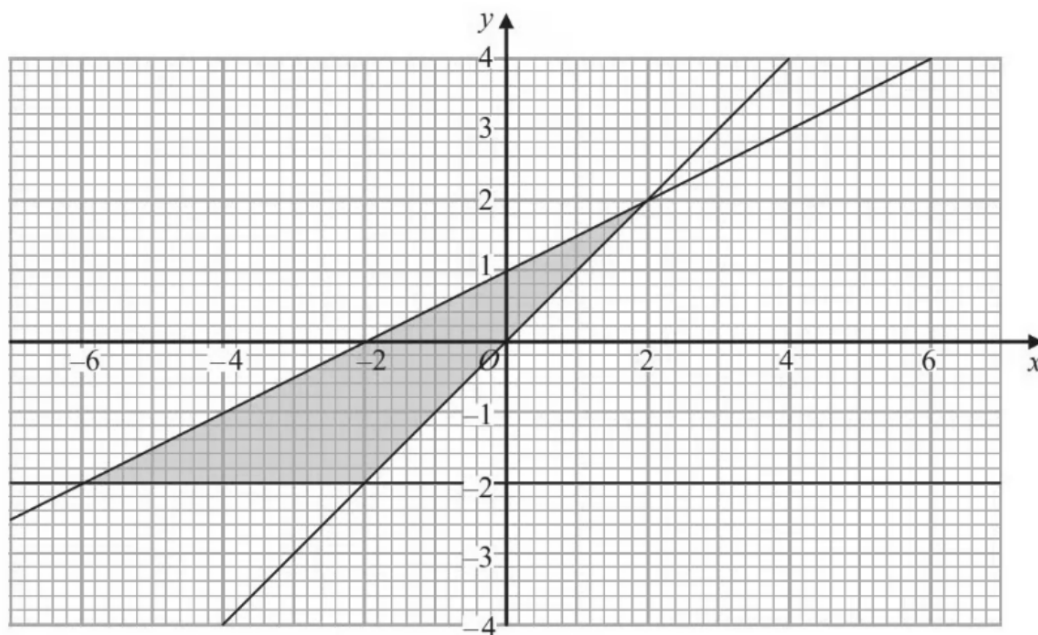
(2 marks)

Q14**PRACTICE**

Practice • Medium

CH 3 • TOPIC 8 4 marks

2



Write down the three inequalities that define the shaded region.

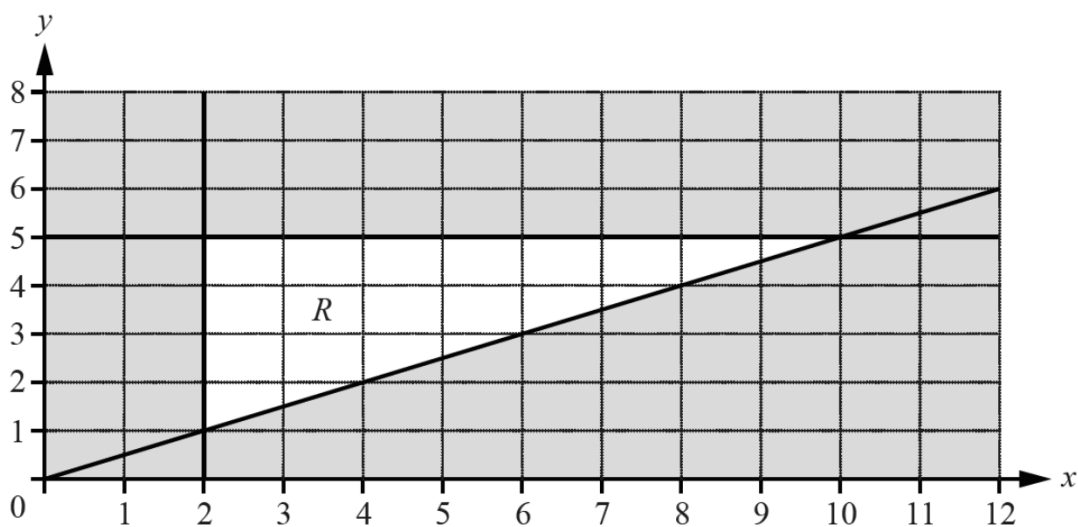
(4 marks)

Q15**PRACTICE**

Practice · Hard

CH 3 · TOPIC 8 6 marks

10 (a)

Find the three inequalities that define the region R .

(4 marks)

- (b) Find the point (x, y) , with integer co-ordinates, inside the region R such that $3x + 5y = 35$

(..... ,)

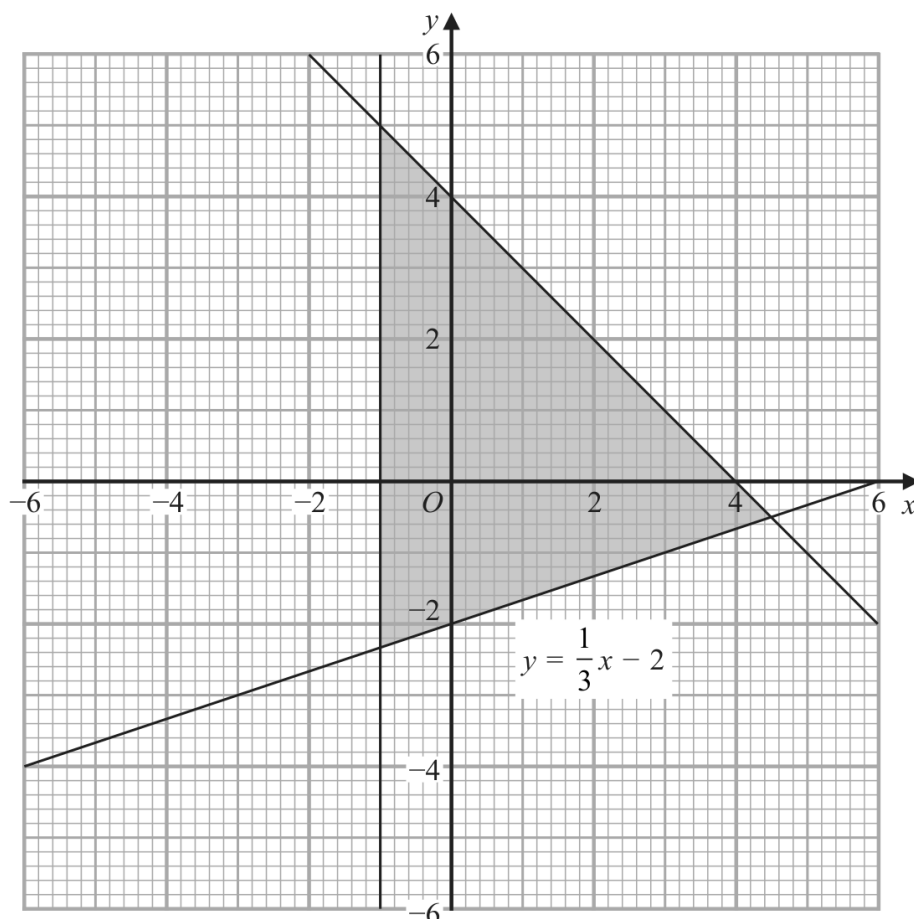
(2 marks)

PAST PAPERS

Q16 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1H · Q10

CH 3 · TOPIC 8 3 marks

- 10 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

.....
.....
.....

(Total for Question 10 is 3 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q11

CH 3 · TOPIC 8 4 marks

- 11 (a) Write down the integer values of x that satisfy the inequality $-2 < x \leq 4$
-

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

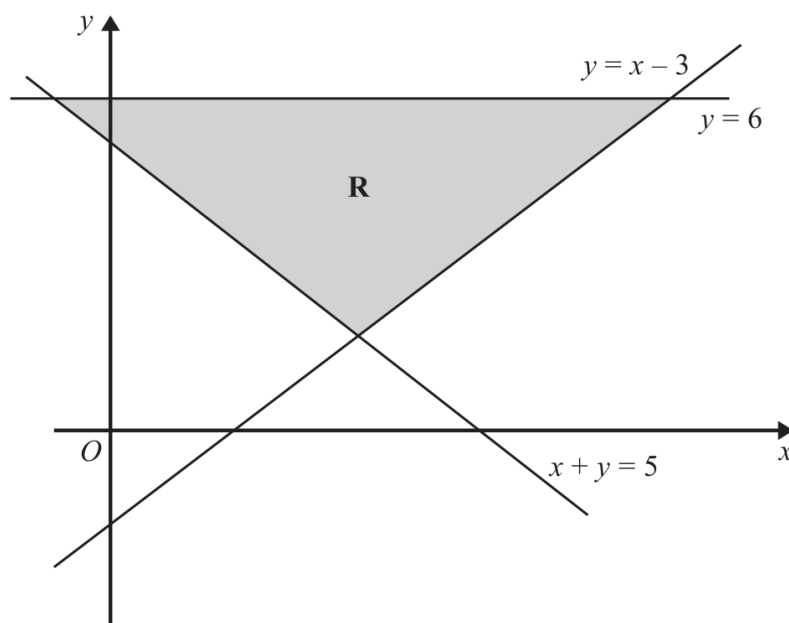


Diagram **NOT**
accurately drawn

- (b) Write down the three inequalities that define the region **R**.
-
-
-

(Total for Question 11 is 4 marks)

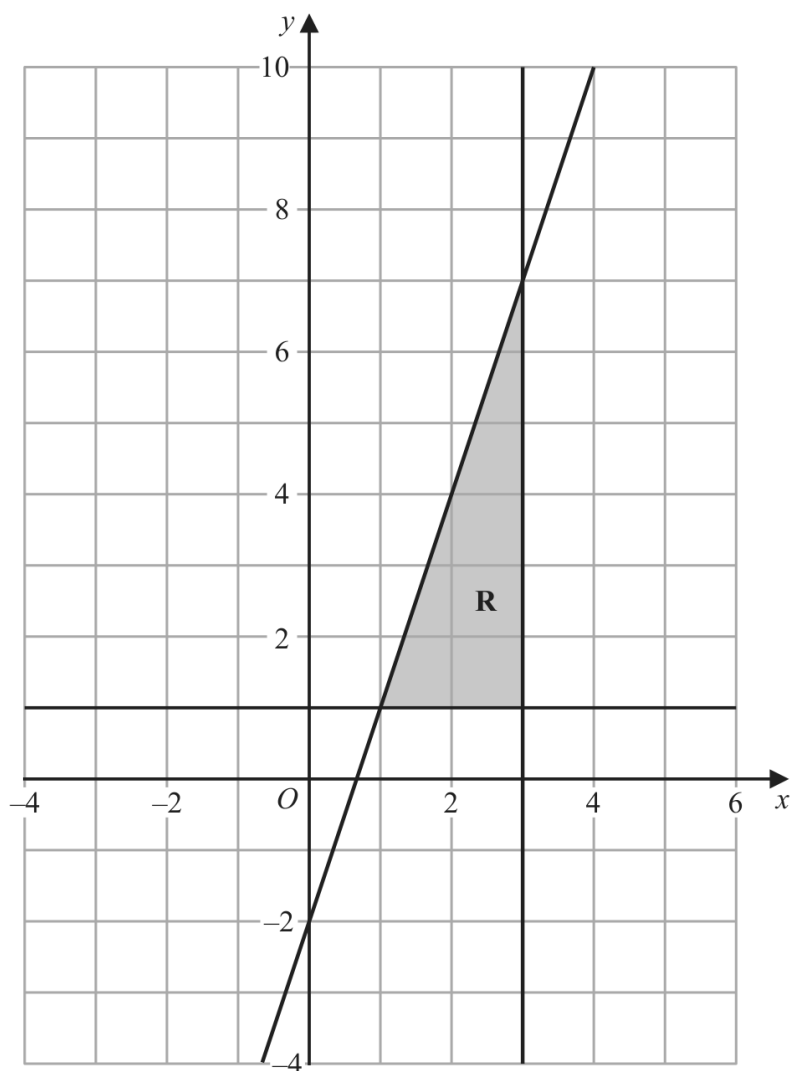
Q18**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q7

CH 3 · TOPIC 8 3 marks

- 7 The shaded region **R**, shown in the diagram below, is bounded by the straight line with equation $y = 3x - 2$ and by two other straight lines.

Write down the three inequalities that define region **R**.



.....
.....
.....

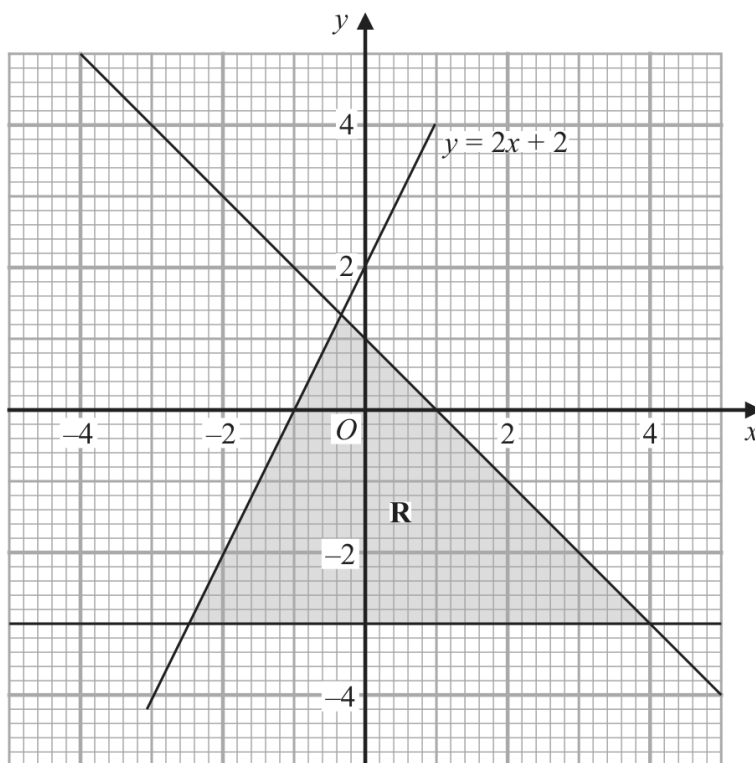
(Total for Question 7 is 3 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q13

CH 3 · TOPIC 8 3 marks

13



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Write down the three inequalities that define **R**.

.....
.....
.....

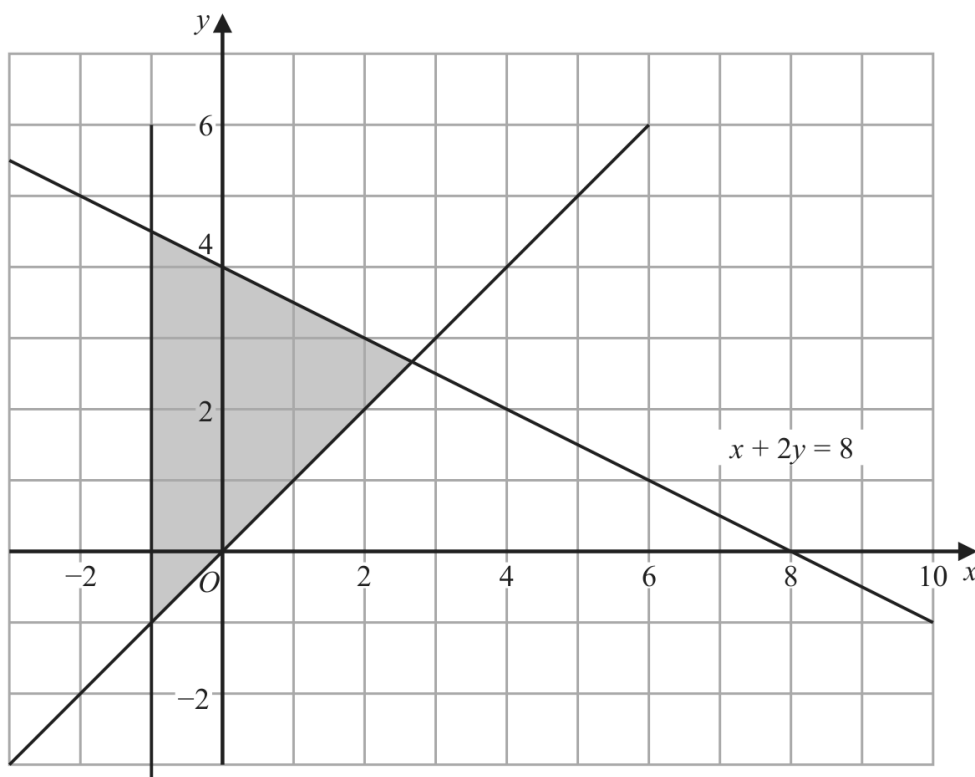
(Total for Question 13 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2H · Q13

CH 3 · TOPIC 8 3 marks

- 13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



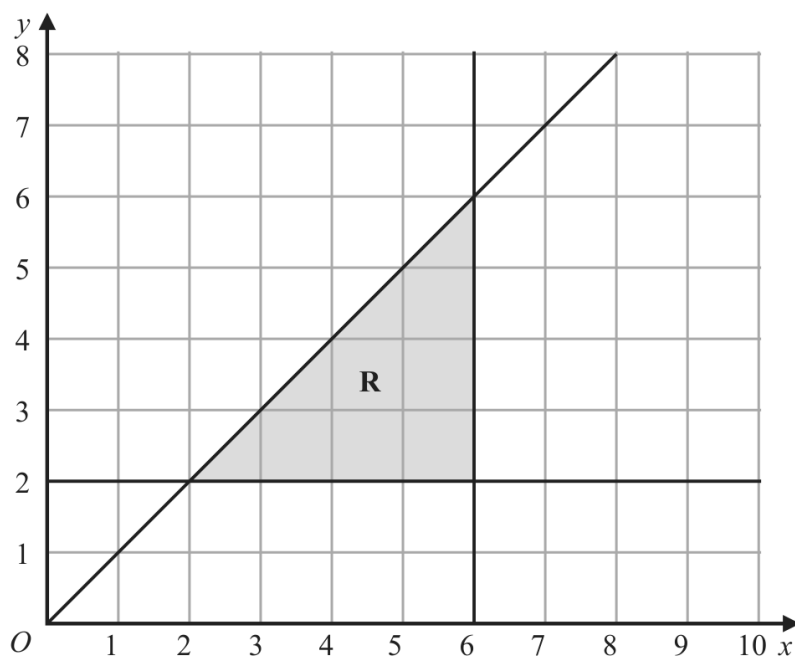
Write down three inequalities that define the shaded region.

.....
.....
.....

(Total for Question 13 is 3 marks)

Q21 PAST PAPER Edexcel 4MA1 • June 2023 • Paper 2HR • Q8

CH 3 • TOPIC 8 5 marks

8 (a) Solve $9 - 4x > 17$ (b) Write down the three inequalities that represent the shaded region **R**

(Total for Question 8 is 5 marks)

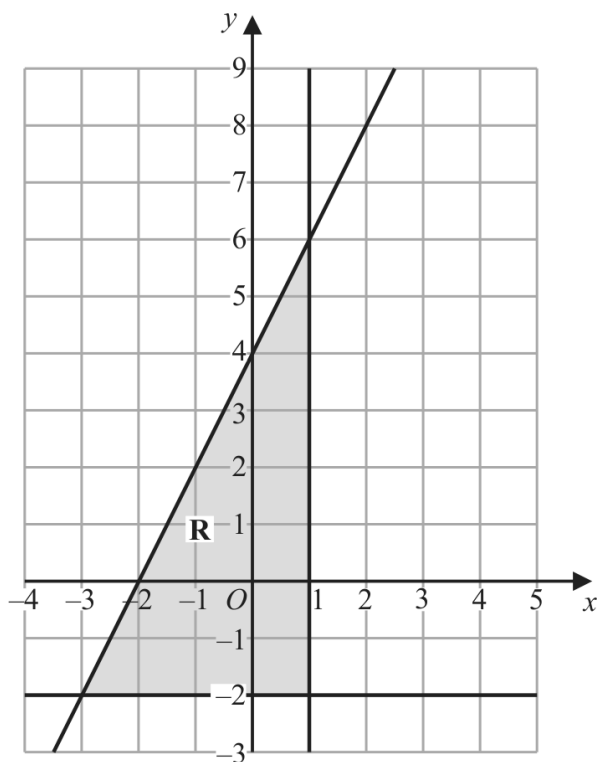
Q22

PAST PAPER

Edexcel 4MA1 • May 2023 • Paper 1H • Q7

CH 3 • TOPIC 8 4 marks

7



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

.....

(Total for Question 7 is 4 marks)

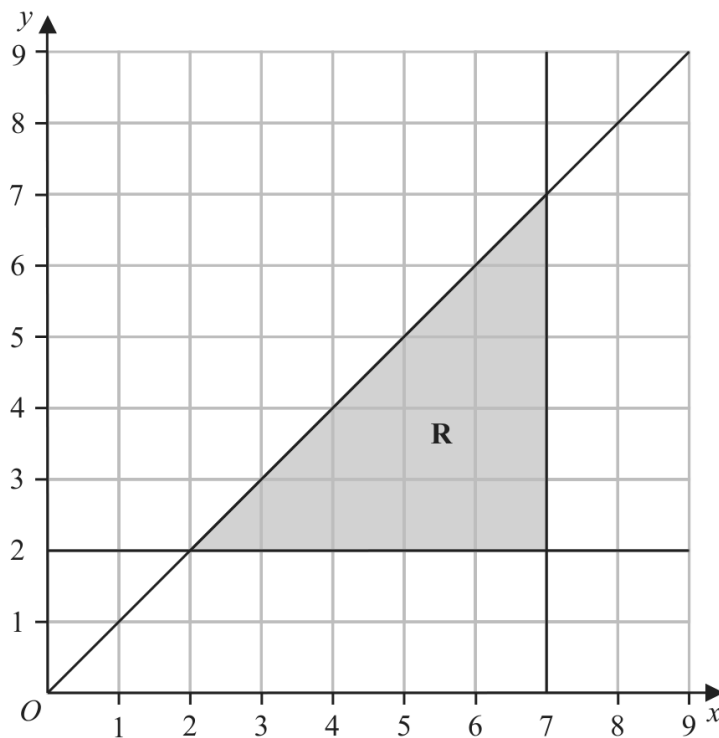
Q23**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q8

CH 3 · TOPIC 8 5 marks

- 8 (a) Solve the inequality $8x - 4 \geq 3x - 10$

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

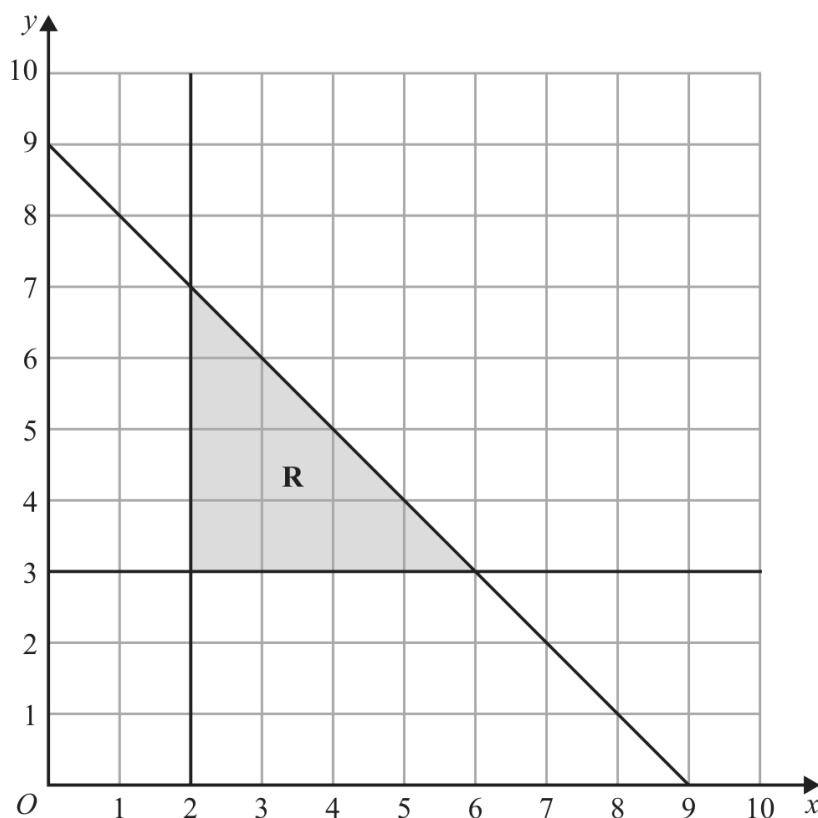
(Total for Question 8 is 5 marks)

Q24 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2H • Q9

CH 3 • TOPIC 8 5 marks

- 9 (a) Solve the inequality $7 - 3t < 2t + 15$

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down three inequalities that define the region **R**

.....
.....
.....

(Total for Question 9 is 5 marks)

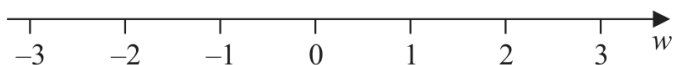
Q25

PAST PAPER

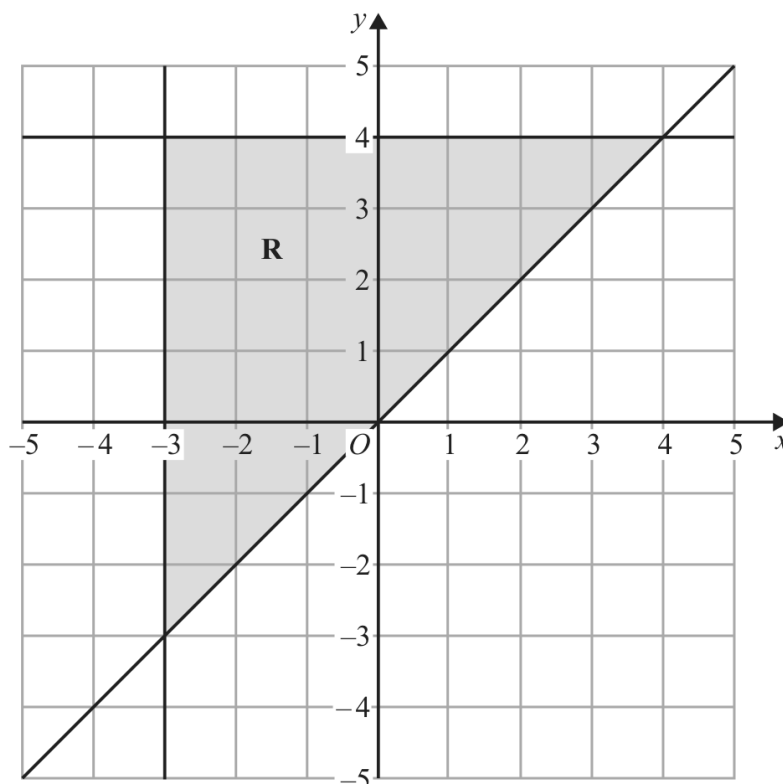
Edexcel 4MA1 • November 2025 • Paper 2H • Q10

CH 3 • TOPIC 8 4 marks

- 10 (a) On the number line, represent the inequality $w < 1$



The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....

(Total for Question 10 is 4 marks)

03 Integer Points in a Region

List the whole-number x values allowed, then for each one the whole-number y values. Check every point against every inequality — strict signs exclude the boundary.

WHEN YOU SEE → YOU DO

“mark each point with integer coordinates”

→ go column by column

“ $3 < x < 7$ and $x + y = 13$ ”

→ list x , find y , check y

a dashed boundary

→ points on it do not count

WORKED MODEL • INTEGER POINTS

Integers with $y > x - 2$, $x + y < 10$, $x > 3$.

1 $x = 4$: $2 < y < 6 \rightarrow y = 3, 4, 5$

column 4

2 $x = 5$: $3 < y < 5 \rightarrow y = 4$; $x = 6$: none

columns 5, 6

ANSWER $(4, 3), (4, 4), (4, 5), (5, 4)$

IDEA 03

Practise: Integer Points in a Region

My questions first, then selected practice.

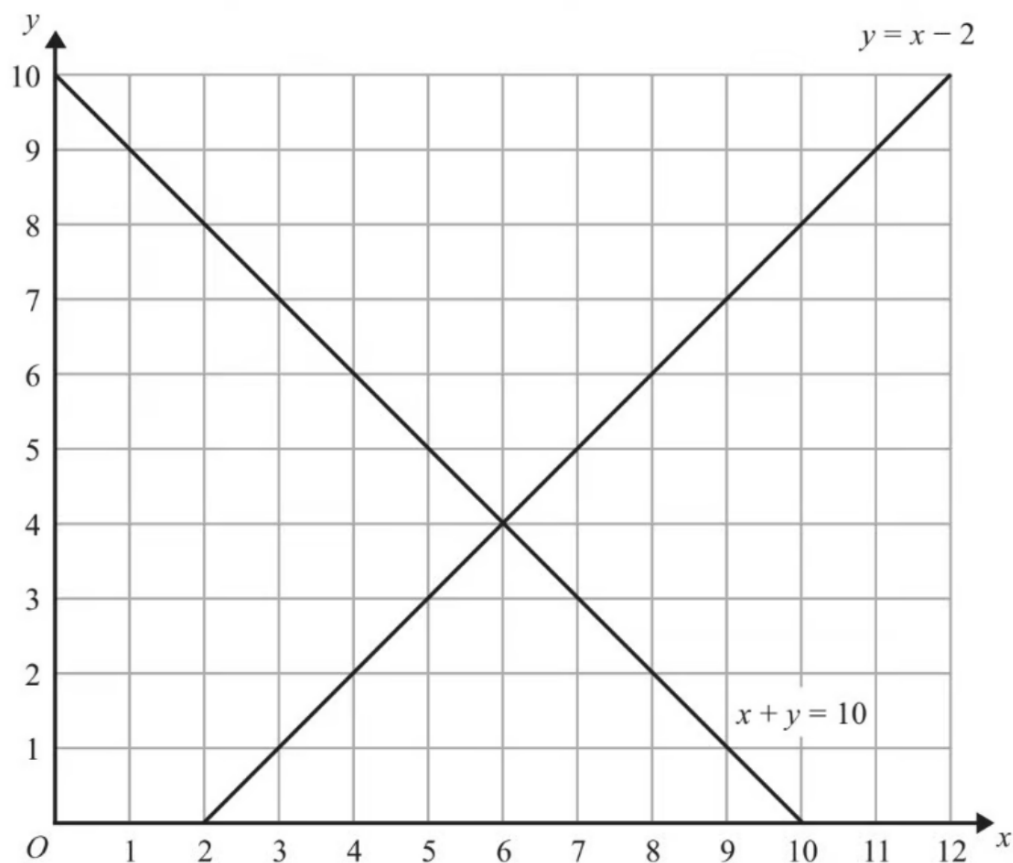
SELECTED PRACTICE

Q26**PRACTICE**

Practice · Medium

CH 3 · TOPIC 8 3 marks

- 1 The lines $y = x - 2$ and $x + y = 10$ are drawn on the grid.



On the grid, mark with a cross (x) each of the points with integer coordinates that are in the region defined by

$$y > x - 2$$

$$x + y < 10$$

$$x > 3$$

(3 marks)

Q27**PRACTICE**

Practice · Hard

CH 3 · TOPIC 8 6 marks

1 (a) Given that x and y are integers such that

$$3 < x < 7$$

$$4 < y < 9$$

$$\text{and } x + y = 13$$

find all the possible values of x .

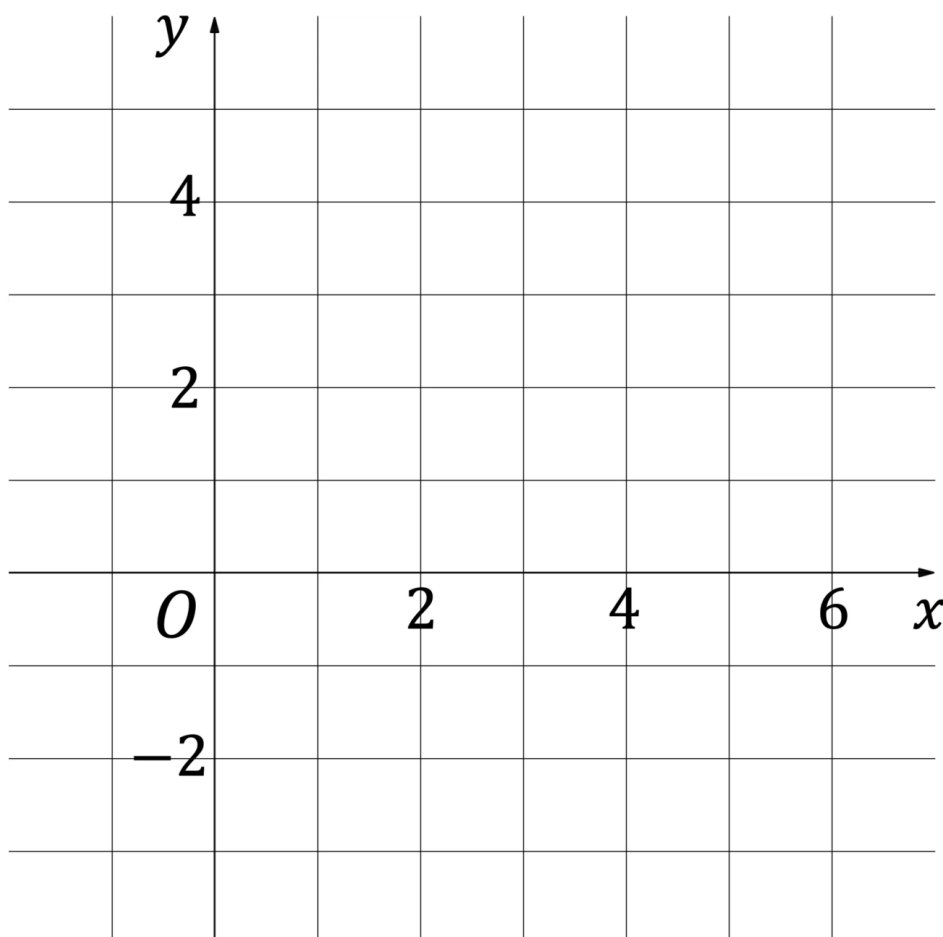
(2 marks)

(b) On the grid below show, by shading, the region defined by the inequalities

$$y \geq -1$$

$$y \leq 4 - x$$

$$y \leq 3x - 1$$

Mark this region with the letter R .

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------|--|
| 01 | Shade | Draw lines (solid $\leq$, dashed $<$), test a point, keep the overlap, label R . |
| 02 | Write | One inequality per line; equation of the line, then the sign from a test point. |
| 03 | Integers | Column by column; strict signs exclude the line. |

BOTTOM LINE Line $\rightarrow$ solid or dashed $\rightarrow$ test a point $\rightarrow$ the region is where all are true.

Answers

Graphing Inequalities • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SHADE THE REGION FROM THE INEQUALITIES

- | | | |
|------------|--|--|
| Q1 | Dr Eslam • Shade | $(1, 2), (4, 2), (1, 5)$ |
| Q2 | Practice • Easy | triangle $(-2, -2), (4, -2), (4, 4)$ |
| Q3 | Practice • Easy | between $x = 2$ and $x = 3$, above $y = 4 - x$,
below $y = 2x + 1$ |
| Q4 | Practice • Medium | draw $y = 1$ (dashed): above $y = 1$, below $y = 2x + 1$, above $7x + 4y = 28$ |
| Q5 | Practice • Hard | draw $y = 3$ (dashed), $y = x$, $x + y = 9$: above $y = 3$, left of $y = x$, below $x + y = 9$ |
| Q6 | Edexcel 4MA1 • January 2021 • Paper 2H • Q4 | lines drawn; R: triangle $(1.5, 1.5), (3, 3), (1.5, 4.5)$ |
| Q7 | Edexcel 4MA1 • January 2022 • Paper 2HR • Q7 | (a) $y = 5 - 3x$ (b) triangle $(1, 2), (6, 2), (6, 7)$ |
| Q8 | Edexcel 4MA1 • June 2022 • Paper 2H • Q2 | triangle $(2, 1), (6, 1), (2, 5)$ |
| Q9 | Edexcel 4MA1 • June 2024 • Paper 2H • Q2 | triangle $(1, 2), (6, 2), (6, 7)$ |
| Q10 | Edexcel 4MA1 • November 2024 • Paper 2H • Q7 | triangle $(3, 1), (6, 1), (3, 4)$ |

IDEA 02 • WRITE THE INEQUALITIES FOR A SHADED REGION

- Q11** Dr Eslam • Write $y \geq 0, x \leq 4, y \leq x$
- Q12** Practice • Easy $0 \leq x \leq 2, 1 \leq y \leq 3$
- Q13** Practice • Easy $a = -1, b = 5, c = 1, d = 4$
- Q14** Practice • Medium $y \geq -2, y \leq \frac{1}{2}x + 1, y \geq x$
- Q15** Practice • Hard (a) $x \geq 2, y \leq 5, y \geq \frac{1}{2}x$ (b) (5, 4)
- Q16** Edexcel 4MA1 • January 2020 • Paper 1H • Q10 $x \geq -1, y \geq \frac{1}{3}x - 2, x + y \leq 4$
- Q17** Edexcel 4MA1 • January 2020 • Paper 2HR • Q11 (a) -1, 0, 1, 2, 3, 4 (b) $y \leq 6, x + y \geq 5, y \geq x - 3$
- Q18** Edexcel 4MA1 • January 2021 • Paper 2HR • Q7 $y \geq 1, x \leq 3, y \leq 3x - 2$
- Q19** Edexcel 4MA1 • November 2021 • Paper 2H • Q13 $y \geq -3, y \leq 2x + 2, x + y \leq 1$
- Q20** Edexcel 4MA1 • January 2022 • Paper 2H • Q13 $x \geq -1, y \geq x, x + 2y \leq 8$
- Q21** Edexcel 4MA1 • June 2023 • Paper 2HR • Q8 (a) $x < -2$ (b) $y \geq 2, x \leq 6, y \leq x$
- Q22** Edexcel 4MA1 • May 2023 • Paper 1H • Q7 $x \leq 1, y \geq -2, y \leq 2x + 4$
- Q23** Edexcel 4MA1 • November 2023 • Paper 1H • Q8 (a) $x \geq -1.2$ (b) $y \geq 2, x \leq 7, y \leq x$
- Q24** Edexcel 4MA1 • June 2025 • Paper 2H • Q9 (a) $t > -1.6$ (b) $x \geq 2, y \geq 3, x + y \leq 9$
- Q25** Edexcel 4MA1 • November 2025 • Paper 2H • Q10 (a) open circle at 1, arrow left (b) $x \geq -3, y \leq 4, y \geq x$

IDEA 03 • INTEGER POINTS IN A REGION

Q26 Practice · Medium $(4, 3), (4, 4), (4, 5), (5, 4)$ **Q27** Practice · Hard(a) $x = 5$ or 6 (b) triangle below $y = 4 - x$ and $y = 3x - 1$, above $y = -1$